

EDUCATION

Swarthmore College — Philadelphia, PA

Aug 2023—May 2027

B.S. General Engineering (Electrical focus) & B.A. Applied Mathematics | GPA: 3.98/4.00

Relevant coursework: Electronic Circuit Applications, Digital Signal Processing, Computer Vision, Data Structures & Algorithms (C++), Partial Differential Equations (PDEs), Real Analysis, Tensor Decompositions, Numerical PDEs, Thermo-Fluid Mechanics, Electromagnetism, Electrodynamics, Statistical Signal Processing

Teaching Assistant for Electrical Circuit Analysis, Aug 2025—Jan 2026 | NCAA Varsity Soccer, Aug 2023—Jun 2026

RESEARCH AND WORK EXPERIENCE

Naval Research Laboratory, Washington, DC

May 2026—Aug 2026

Plasma Synthesis and Materials Characterization Research Intern

- Developed inductively coupled plasma (ICP) reactor to synthesize nanoparticles; conducted plasma experiments.
- Minimized reflected power in ICP by tuning the RF matching network using Smith chart; mitigated RF EMI.
- Used scanning electron microscopy and energy dispersive spectroscopy to characterize nanomaterial compacts.
- Designed and ran atomic-layer-deposition (ALD) experiments to determine how dose parameters affect ALD coatings.

Advanced Materials Intern & Research Assistant

May 2025—Aug 2025

- Measured thermal diffusivity and specific heat of thermoelectric semiconductors using laser flash analysis (LFA).
- Designed and built Arduino-controlled LN₂ dispenser with closed-loop thermocouple monitoring to automate LFA experiments, including a GUI and data-logging interface to the lab computer.
- Developed experiments to improve the lab’s measurement procedure for IR-transparent and high-resistance samples.

Swarthmore College NaSCaRS Lab – Applied Mathematics Research Assistant, Philadelphia, PA

Jan 2024—present

- Utilizing computational fluid dynamics and numerical methods to develop high-order accurate PDE solvers for kinetic & fluid plasma models in MATLAB.
- Co-authored [paper](#) on low-rank, structure-preserving solver for Vlasov-Fokker-Planck kinetic-ion fluid-electron model.
- Poster presentations: 2025 SIAM NNP conference, 2024 & 2025 Sigma Xi Swarthmore poster sessions.

Swarthmore College Engineering – Electrical Engineering Research Assistant, Philadelphia, PA

Nov 2023—May 2024

- Researched wind-energy harvesting technology and used Arduino, MATLAB & ViscousFlow to simulate air vortex-shedding & electrical induction power output for an oscillatory wind-energy harvester.

Oregon Health and Science University – Software Engineering Intern, Portland, OR

Jun 2022—Aug 2022

- Used Kotlin to develop Android app that detects heart murmurs using audio data from a Bluetooth stethoscope.

PROJECTS

Convolutional Neural Network for Hand-Drawn Image Classification: [project link](#)

May 2026

- Designed and trained a CNN to classify 250 object categories, deployed it online for real-time evaluation using ONNX.

Analog Circuit and PCB Design – Fusion360

Aug 2025—Dec 2025

- Designed and fabricated PCBs combining MOSFETs, sensors, actuators, op-amps, comparators, and audio amplifiers across four projects: [audio equalizer](#), [temp-controlled fan](#), [light-optimizing solar panel](#), [regulated DC power supply](#).

AI Python Stock Trading: trained neural network on technical indicators to trade stocks using Alpaca API, [link](#)

Feb 2025

Generative Adversarial Network (GAN): Image-generating model trained on web-scraped art, [link](#)

Apr 2022—Feb 2023

Technical & Professional Skills

Technical: SPICE, PCB Design, CAD, Fusion360, Soldering, Embedded systems, PDE solvers, Numerical methods

Programming: Python, MATLAB, Arduino, Java, C++, HTML/JavaScript, LaTeX

Software: Git, SolidWorks, AutoCAD, NumPy, TensorFlow, PyTorch, OpenCV

Languages: Spanish (Fluent), Global Seal of Biliiteracy (2022) | Citizenship: USA, Ireland

Honors and Scholarships

Tau Beta Pi Engineering Honors Society

Mar 2026

Delaware Valley Engineers Undergraduate Scholarship, Delaware Valley Engineers Society

Feb 2025, Mar 2026

Donna Prentice Memorial Scholarship, American Society of Civil Engineers

Feb 2024

National Merit Scholarship, National Merit Scholarship Corporation

Apr 2023